

DC-DC CONVERTERS – BUCK CONVERTERS 2 (Design)

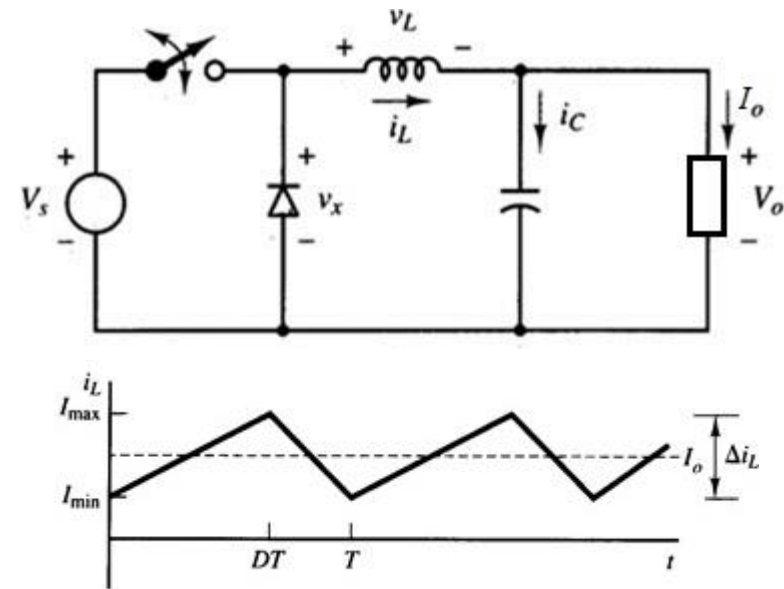
BUCK2-1 Continuous Inductor Current Design

To keep the inductor current continuous, its minimum current must not drop to zero. Let's take another look at the buck converter circuit and inductor current waveform.

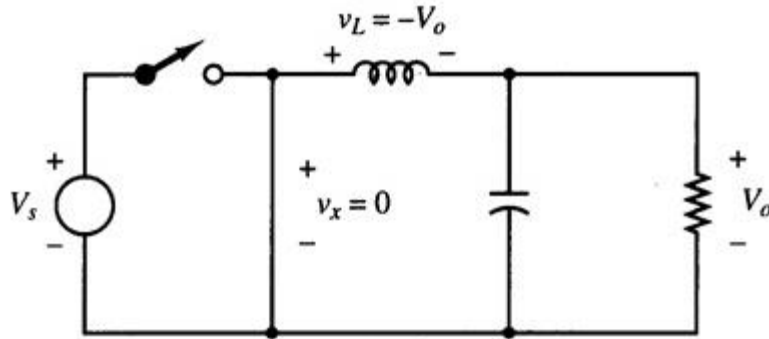
Will become **discontinuous conduction** for inductor when

$$\Delta i_L / 2 > I_o. \text{ As such, by design, we should try to make } I_o > \Delta i_L / 2.$$

We then need to find an expression that tells us what the minimum inductor value is, dependent on the key operational parameters of output voltage, output average current, duty cycle, and switching period (or frequency).



Can do this by calculating Δi_L . Could look at when switch is on, or when it is off. We'll use the off period.



$$\Delta i_{L,open} = -\left(\frac{V_o}{L}\right)(1-D)T \quad (\text{from } v_L = L \frac{di_L}{dt} = -V_o)$$

$$\text{So } |\Delta i_L| = \frac{V_o(1-D)}{Lf_s}$$

Substituting this expression for Δi_L into the previous inequality gives us;

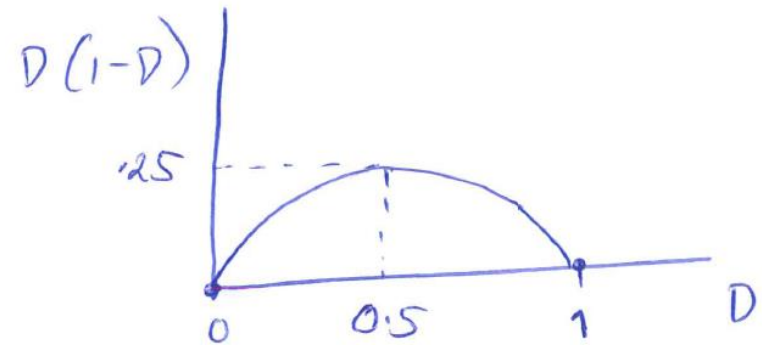
$$I_o > \frac{V_o(1-D)}{2Lf_s} \quad \text{or} \quad L > \frac{V_o(1-D)}{2f_s I_o}$$

But for the Solar Car project, it is the input voltage that is fixed, where $V_o = DV_s$. Making this substitution for V_s gives;

$$L > \frac{V_s D(1-D)}{2f_s I_o} \quad \text{where the worst case is } D = 0.5$$

So

$$L > \frac{V_s}{8f_s I_o}$$



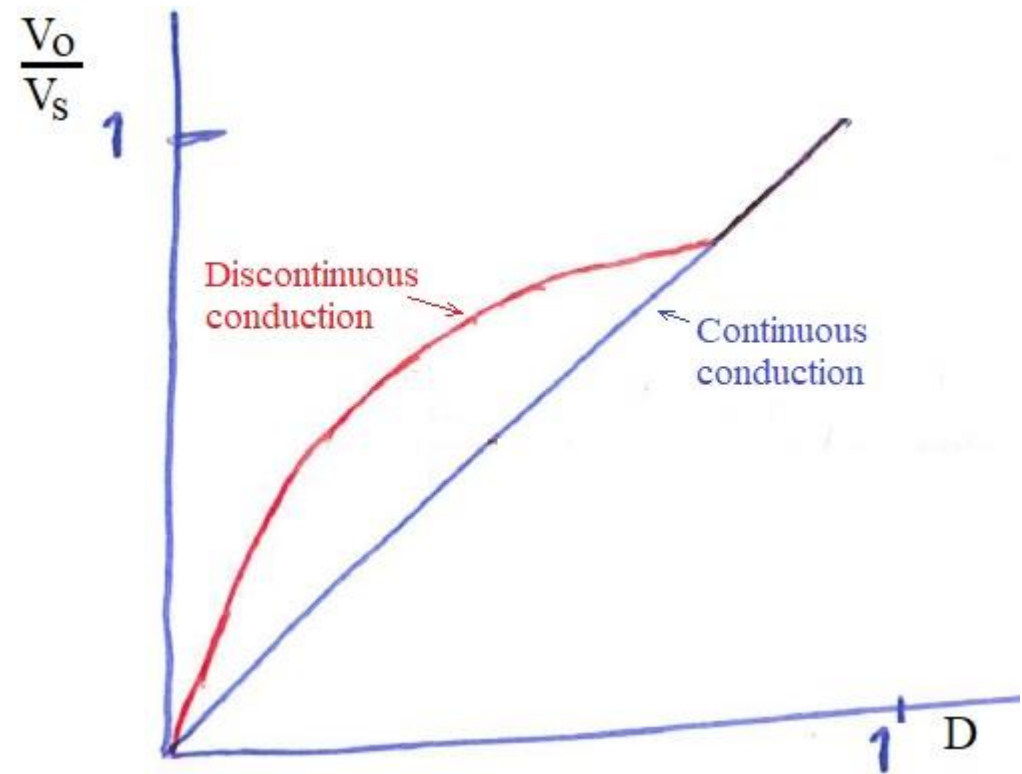
For an application where a value of $D = 0.5$ may be experienced.

Choice of switching frequency depends on many factors. Higher frequencies allow for smaller inductors and capacitors to be used which is advantageous for economic and sustainability reasons. However, higher frequencies introduce more switching and magnetic component losses, and electrical noise which can be disadvantageous for efficiency and noise emissions reasons. Modern converters often work in the hundreds of kilohertz range.

Discontinuous Current (Discontinuous Conduction)

If the inductor current is allowed to become discontinuous, we get;

- Non-linear gain (harder to control)
- Output voltage is dependent on load current, which means D has to change for different loads
- For a given power flow (input to output), since the current is zero at times, the peak supply currents are higher, requiring higher current rated switching devices (which are more expensive and slower switching) and producing more harmonics (more electrical noise and higher losses).



BUCK2-2 Output Voltage Ripple

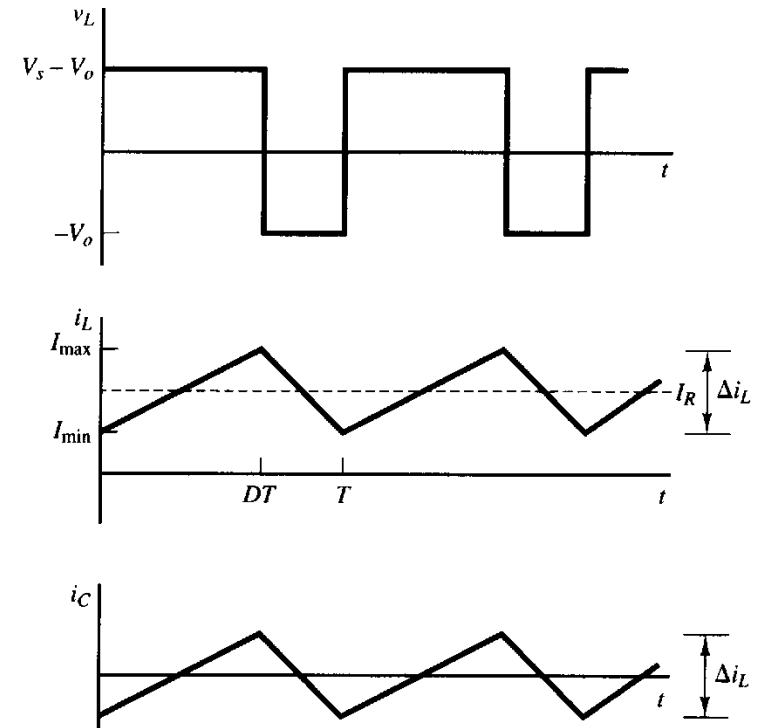
We now consider a more practical situation for the output voltage where there is some ripple, but is small enough not to have an appreciable effect on the previous analysis.

From the figure here, it can be seen that

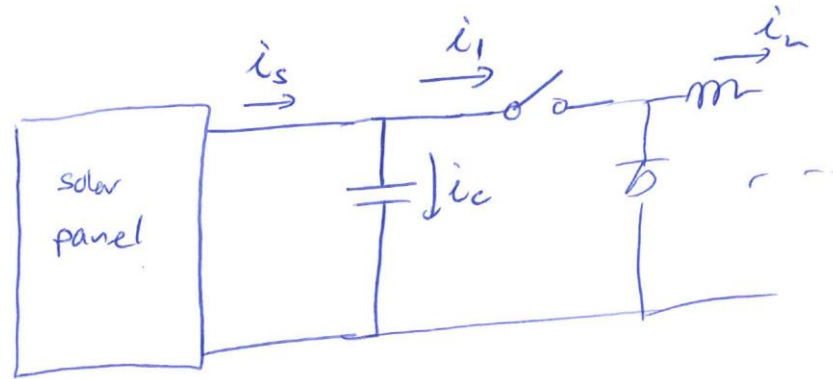
$$i_C = \Delta i_L$$

If there is current flowing in the capacitor, then there must be an associated voltage variation. By consideration of the voltage-charge relationship for capacitors (i.e. $Q=CV$), and $i_C = C \frac{dv}{dt}$ we get,

$$\Delta V_o = V_o \left(\frac{1-D}{8LCf^2} \right). \text{ Many applications aim to have } \Delta V_o \text{ to be less than 1\% of } V_o.$$

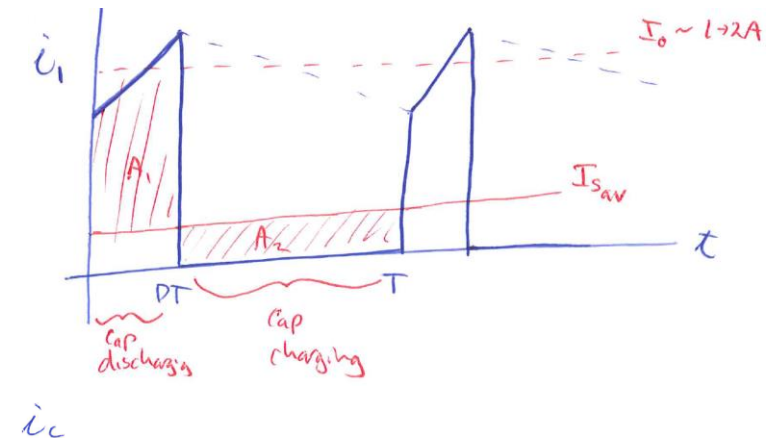


BUCK2-3 Input Capacitance (Solar Car application)



We want to design the converter such that the solar panel current ripple is as small as possible, and also have a low voltage ripple so that it can keep losses low and maintain operation for maximum power output. To do this, we need an appropriately sized input capacitor. For basic sizing, assume i_s has zero ripple.

By KCL, $i_c = i_s - i_1$



Waveforms.

Capacitor voltage ripple (from $i_c = C \frac{\Delta V}{\Delta t}$)

Looking when current is constant, $\Delta V = \frac{I_s(1-D)T}{C}$

But more likely to know I_o .

Relationship between I_s and I_o ?

Basic approximation is the converter is 100% efficient. In this case,

$$I_o V_o = I_s V_s \quad (\text{all DC values})$$

We know $V_o = D V_s$

So

$$I_s = D I_o, \text{ giving } \Delta V_s = \frac{I_o D (1-D)}{f_s C}$$

Worst case (giving largest value of *minimum C*)?

In addition to specific component values, RMS and peak currents and voltages must be considered when choosing the components for constructing converters. For example, a 100V, 5A_p MOSFET wouldn't last long in a buck converter where the source voltage was 300V and peak source current was 20A_p.

Also, for higher switching frequencies and/or power, where there is enough switching and/or conduction loss to be concerned about, thermal control of the switches may be required through the use of heat sinks.

Example: A Buck converter application has the following requirements/specifications;

- Load average DC voltage and current of 8V, and 2A.
- Input average DC voltage of 12V.
- Switching frequency of 80 kHz.
- Output inductor peak-to-peak current ripple less than average output current value.
- Output and input voltage ripple less than 1%.

What is the minimum size of the output inductor, output capacitor and input capacitor?

